# Chapter 6, Problem 4P

## Problem

A voltage across a capacitor is equal to [2 − 2 cos(4*t*)] V and the current flowing through it is equal to 2 sin (4*t*) $\mu$A. Determine the value of the capacitance. Calculate the power being stored by the capacitor.

## Step-by-step solution

**Step 1** of 2

The capacitor current is,

$$i(t) = C\frac{dv(t)}{dt}$$

Substitute $2 - 2\cos(4t)\ \text{V}$ for $v(t)$ and $2\sin(4t)\ \mu\text{A}$ for $i(t)$.

$$\begin{aligned}
2\sin(4t)\times10^{-6} &= C\frac{d}{dt}\left[2-2\cos(4t)\right] \\
2\times10^{-6}\sin(4t) &= C(-2)(-4\sin 4t) \\
\frac{2\times10^{-6}\sin(4t)}{8\sin 4t} &= C \\
\frac{10^{-6}}{4} &= C
\end{aligned}$$

$$\begin{aligned}
C &= 0.25\times10^{-6} \\
&= 0.25\ \mu\text{F}
\end{aligned}$$

Thus, the value of the capacitor is $\boxed{0.25\ \mu\text{F}}$.

**Step 2** of 2

The expression for the power is,

$$p(t) = v(t)i(t)$$

Substitute $2 - 2\cos(4t)\ \text{V}$ for $v(t)$ and $2\sin(4t)\ \mu\text{A}$ for $i(t)$.

$$\begin{aligned}
p(t) &= \left(2-2\cos(4t)\right)\left(2\sin(4t)\times10^{-6}\right) \\
&= 4\left(1-\cos(4t)\right)\sin(4t)\ \mu\text{W}
\end{aligned}$$

Thus, the power being stored in the capacitor is $\boxed{4\left(1-\cos(4t)\right)\sin(4t)\ \mu\text{W}}$.